

Month 1 and 2

Reading literature and lectures regarding drone Modelling and its implementation

Material 1: Literature Study

I started the research project by performing a thorough study of the research paper "Mathematical Model of a Multi-Rotor Drone Prototype" by M.A. Stamate, A.F. Nicolescu, and C. Pupază, published in Proceedings in Manufacturing Systems, Vol. 12, Issue 3 (2017).

The paper covers the mathematical modeling, design, and development of a multi-rotor aerial vehicle. It emphasizes deriving the equations needed to predict the drone's behavior under various flight conditions. The authors discuss the mechanical setup, propulsion system, avionics integration, and how they validate the design using simulation tools before implementing it physically. The study details different flight maneuvers such as hovering, ascending, forward flight, lateral motion, and rotation about the vertical axis. It shows how mathematical models can analyze and predict the drone's dynamic response.

From this literature review, the following concepts were understood:

- Basics of multi-rotor UAV design ,operation and the parameters involved.
 - Body-fixed and inertial coordinate systems used to describe drone movement.
 - Six Degrees of Freedom (6-DOF) linked to translational and rotational motion.
 - Forces and moments produced by the propellers and their impact on vehicle stability and maneuverability.
 - Introduction to Newton-Euler equations.
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- The importance of precise mathematical modeling as a base for simulation, controller design, and autonomous flight.

Video 2: Quadcopter System Architecture and Control Loop

▶ how to create a mathematical model of a Quadcopter

This lecture introduced the overall setup of a quadcopter system and explained how different subsystems work together during flight. We studied the complete feedback control loop, starting with the user or mission input. This was followed by the controller, motor commands, quadcopter dynamics, onboard sensors, and sensor fusion, which then sent the measured state back to the controller. We also discussed the role of various onboard sensors, including the IMU, GPS, pressure sensor, optical sensors, camera, LiDAR, and radio modules.

The lecture also presented a block diagram of a mathematical quadcopter model. It showed how physical factors such as propeller thrust, aerodynamic forces, torque, moment of inertia, vehicle mass, center of gravity, and quadcopter configuration are expressed mathematically. We used modeling approaches like state-space representation and Laplace transforms. This helped us understand the system-level workflow that forms the basis of quadcopter modeling and simulation.

Video 3: Momentum Theory and Aerodynamic Principles

▶ Quadrotor Equations of Motion and Control KCC Final 4 2023 Video

This lecture explained the aerodynamic principles that govern quadcopter flight using momentum theory. It covered topics like airflow through the propeller disc, thrust generation, induced velocity, blade structure, and how propeller aerodynamics affect flight performance. Although we didn't directly include these concepts in the mathematical

modeling and simulation developed during the internship, they offered a solid understanding of rotorcraft aerodynamics. This knowledge will be useful for future work on propeller selection, performance analysis, experimental validation, and real-world flight testing.

Video 4: Derivation of the Equations of Motion for a Quadcopter

 [Quadrotor Equations of Motion and Control KCC Final 4 2023 Video](#)

The equations of motion of the quadcopter were derived in detail in this lecture using the Newton-Euler and Lagrangian methods. The derivation started from the basics of rigid body dynamics and progressively derived the equations of translational and rotational motion for the quadcopter.

The lecture introduced topics like inertial frame and body-fixed frame, rotation matrices, Euler angles, angular velocity, moments of inertia, force/moment balances, and the derivation of the full 6-degrees-of-freedom (6-DOF) dynamic model of the quadcopter. The differences and similarities of the Newton-Euler and Lagrangian method were also covered, showing how both methods result in the same equations of motion of the quadcopter.

By studying the derivation, the physical meaning behind every equation of motion was clearly understood instead of just taking them as known equations without knowing their physical meaning. This was extremely helpful when building the quadcopter model in MATLAB since all state variables, parameters, forces, moments, and matrices used in the model could now be understood starting from scratch.

Video 5: MATLAB Modelling and Simulation of Quadcopter Dynamics

 [\[AE450 Lec10 -Da\] MATLAB Simulation of a Quadrotor UAV Dynamics and Control](#)

This lecture was about using MATLAB to make a model of a quadcopter. I used a kind of programming called object-oriented programming to make this model. The equations that describe how the quadcopter moves were turned into code. This code can predict how the quadcopter will move in the air.

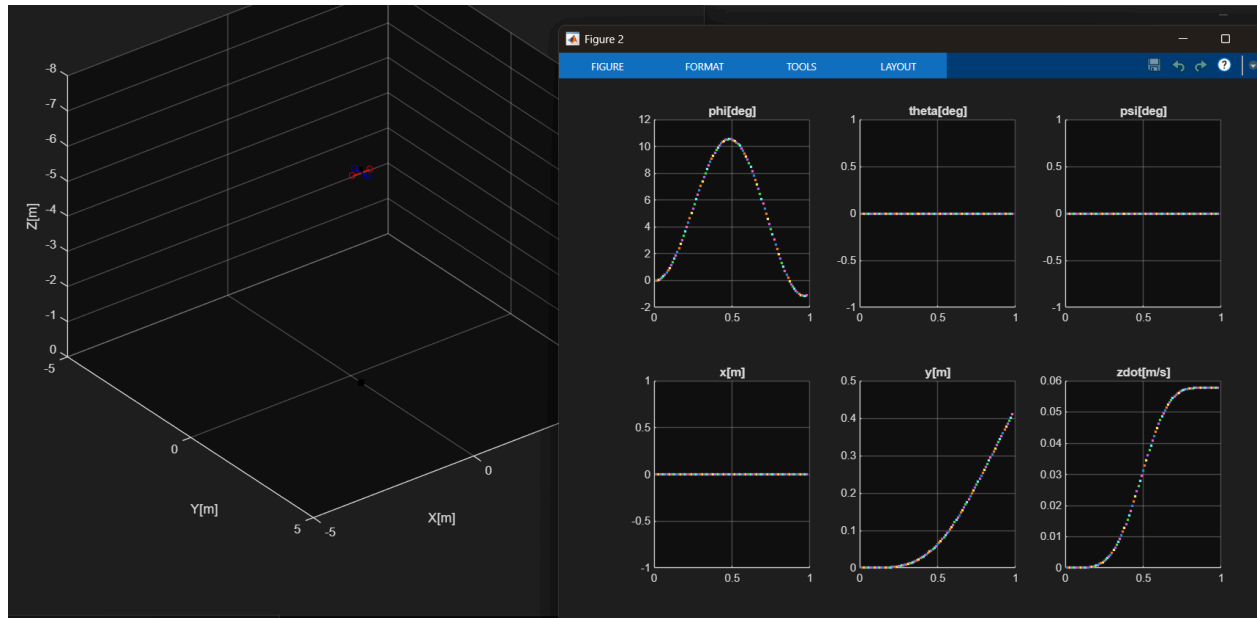
First I defined the properties of the quadcopter like its mass and size. I also set up the conditions, like where the quadcopter starts and how fast it is going. I made a class to hold all the information about the quadcopter, like its properties and how it is controlled. This made it easy to update the model as the simulation ran.

The main part of the model was based on some equations that describe how the quadcopter moves. These equations take into account things like the force of the motors, the weight of the quadcopter and how it is spinning. At each step of the simulation the model calculated how the quadcopter was accelerating and then used that to update its position and speed. We used math to transform the coordinates of the quadcopter from its own frame of reference to the world's frame of reference.

We also added a control system to the model. This system used a kind of control called PID control to make the quadcopter follow a desired path. The control system looked at the difference between where the quadcopter was and where it was supposed to be and then adjusted the motors to make it go to the place.

The simulation showed us what the quadcopter was doing in time. We could see an animation of the quadcopter moving around and we could also look at plots of how its position and speed were changing over time. This allowed us to see how the quadcopter responded to control inputs and to adjust the model to make it more realistic.

The finished model was a tool for understanding how quadcopters work. We could use it to test control strategies and see how they affected the stability and performance of the quadcopter. This could be really useful for designing and building quadcopters and for making sure they are safe and reliable. The quadcopter model was a help in understanding UAV modelling and attitude control. The quadcopter simulation was a part of the lecture. The quadcopter model was used to study the effect of control strategies on the quadcopter.



These are the simulation windows. 6 different parameters are plotted in real life and the motion of the drone is also simulated. The values can be later used to train drone performances.